

On the Limitations of Metrics for Quantum Circuit Execution

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Abstract—Quantum-computing papers routinely report “metrics” of execution success: success rate, Hellinger fidelity, total variation distance, process-spectrum tomography, expected success probability, and a growing set of derived scores. These quantities do not measure the same attribute, do not live on the same domain, and do not permit the same operations. The mismatch is usually harmless for a single job, a fixed reference, and a correctly transformed threshold: strictly monotone pairs such as total variation distance and its complement produce the same pass/fail decision. The mismatch becomes decision-relevant under aggregation across jobs, composition of subsystem errors, ratio or interval claims, cross-task comparison, and missing reference information. This paper applies representational measurement theory and the Goal-Question-Metric approach to job-level Sampler and Estimator quantities. It separates distance metrics from similarities and unary scores, introduces four reference classes that decide whether a formula is even computable, and gives a short decision guide. A case study of the Differential Success Rate profile shows that two of its entries are distances on a coarse simplex, while the remaining entries are scores with stated limitations.

Index Terms—Quantum software measurement, representational measurement theory, Goal-Question-Metric, Hellinger fidelity, total variation distance, Sampler, Estimator, Differential Success Rate

I. INTRODUCTION

A developer has shots from a quantum processing unit (QPU) and must decide whether the execution succeeded. Hardware returns bitstrings. Those bitstrings do not determine the attribute being measured. The first distinction is whether the shots are interpreted as a Sampler histogram or known-answer set, or as an Estimator contraction against observables. Both start from the same multiset of outcomes. They measure different attributes of that entity [1]. Applying a Sampler formula to an Estimator result, or the reverse, is a change of attribute, not a change of name.

Quantum-information papers already establish that fidelity is not a distance and that several induced distances are metrics [2]–[5]. That fact is not the contribution of this paper. The remaining problem in software-engineering practice is different: execution reports treat success rate, Hellinger fidelity (HF), total variation distance (TVD), process-spectrum tomography (PST), expected success probability (ESP), and contrast scores as interchangeable quantities, then average them, threshold them at 0.8, or compare them across tasks that do not share a reference.

If two quantities are strictly monotone transforms of one another, then for one job, one fixed reference, and a converted

threshold they produce the same decision. $\text{TVD} \leq 0.2$ is the same single-job call as $1 - \text{TVD} \geq 0.8$. Hellinger distance and Hellinger fidelity are likewise equivalent per job after threshold conversion. Calling one of them a metric and the other a fidelity does not, by itself, change that call.

The distinction becomes decision-relevant outside that setting:

- **Aggregation.** The arithmetic mean of a nonlinear transform need not preserve a ranking obtained from the mean of the original quantity.
- **Composition.** The triangle inequality licences bounds on concatenated errors. A similarity does not inherit that licence until it is mapped back to a distance.
- **Scale.** Ratio statements and universal cutoffs require a defensible scale type [1].
- **Reference.** A formula that is a metric on paper is inapplicable when its reference information cannot be obtained.

The contributions of this paper are:

- 1) an analysis of when the choice of quantity leaves a single-job decision unchanged and when it can change a conclusion;
- 2) a classification of job-level Sampler and Estimator quantities using representational measurement theory and GQM, including four reference classes;
- 3) a decision procedure whose first question is Sampler versus Estimator;
- 4) a case study of the Differential Success Rate (DSR) profile as a known-answer record rather than a single metric.

Two tests for the word “metric” should be kept separate. The first is mathematical: axioms M1–M4 on a named space. The second is representational: a valid measure of a named attribute of a named entity [1], [6]. A quantity can pass one test and fail the other. Hellinger distance passes both as a measure of histogram closeness, yet it can still be the wrong attribute if the developer only needed mass on an answer set E . Chance-corrected success can be a valid ordinal score of above-chance mass and still fail as a distance. ESP can be a useful prediction and still fail as a measure of the observed job.

This work is independent of any conference proposal of DSR as a headline metric. It does not introduce a new process distance and does not claim quantum advantage. DSR appears only as a case study after the classification is in place.

The remainder of the paper is organised as follows. Section II reviews distance axioms and measurement theory. Section III discusses related work. Section IV defines the execution model and reference classes. Section V classifies

common quantities. Section VI states the analytical results. Section VII reports the evaluation. Section VIII discusses selection of a quantity and the DSR case study. Section IX lists threats to validity. Section X concludes.

II. BACKGROUND

A. Distance Metrics and Related Quantities

The classical Hellinger distance and total variation distance on a finite simplex Δ_n are

$$H(P, Q) = \sqrt{1 - \sum_x \sqrt{P(x)Q(x)}} = \sqrt{1 - \text{BC}(P, Q)}, \quad (1)$$

$$\text{TVD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|. \quad (2)$$

Qiskit's Hellinger fidelity is the square of the Bhattacharyya coefficient, $F_H(P, Q) = \text{BC}(P, Q)^2 = (1 - H(P, Q)^2)^2$, not $1 - H$. Both H and TVD are metrics on Δ_n as the restrictions of the L^2 and L^1 norms after the usual embeddings [5], [7]. This paper uses those standard facts; it does not reprove them.

On a set X , a function $d : X \times X \rightarrow \mathbb{R}$ is a *distance metric* if and only if it satisfies

- **M1** $d(x, y) \geq 0$,
- **M2** $d(x, y) = 0 \iff x = y$,
- **M3** $d(x, y) = d(y, x)$,
- **M4** $d(x, z) \leq d(x, y) + d(y, z)$.

Related objects are not candidates for the same test. A *pseudometric* relaxes M2 to $d(x, x) = 0$. A *quasimetric* drops M3. A *divergence* such as Kullback–Leibler is typically neither symmetric nor a triangle function. A *similarity* or *fidelity* maps equality to 1 and decreases as the objects separate; it is not a distance function. A *score* is unary: one job plus a goal or model. M1–M4 are not defined for it.

Two conventions follow. First, $1 - d$ is not itself a metric even when d is. Second, a similarity cannot fail the triangle inequality, because that axiom is not defined for that object. If an induced dissimilarity $1 - F_H$ is examined, the claim must be about $1 - F_H$, not about F_H . Standard completions of Bhattacharyya overlap are the Hellinger distance $H = \sqrt{1 - \text{BC}}$ and the angle $\arccos \text{BC}$ [3], [7]. This paper reserves *distance metric* for M1–M4 on a named space and calls everything else a measure, score, or similarity only after the attribute is named.

B. Representational Measurement Theory

Fenton and Bieman [1] treat measurement as a mapping from an empirical relational system to a numerical one that preserves the intended relations (the representation condition). A job is not itself the measured object. The measured object is a named attribute of a named entity: histogram closeness to a reference, mass on E , peak dominance, or deviation of an expectation.

Five classical scale types restrict which statements are meaningful.

- **Nominal.** Labels only. A provider name is not a success quantity.

- **Ordinal.** Order is preserved; differences are not. A ranking of jobs by F_H supports “closer than,” not “twice as close.”
- **Interval.** Differences are meaningful; the zero is conventional. Pauli expectations on $[-1, 1]$ are closer to this type than to a success rate.
- **Ratio.** A non-arbitrary zero licences statements such as “twice as far.” Distances that satisfy M1–M4 are the usual candidates; $1 - d$ is not automatically one of them.
- **Absolute.** Counts of shots sit closest to this type. Derived proportions inherit meaning only after the derivation is stated [1].

Most published fidelities are *derived* measures: they are functions of counts, a reference, and a modelling choice. Fenton requires a meaningfulness argument for such calculations, not only a formula with values in $[0, 1]$.

Fenton [6], [8] warned software engineering against calling every number a metric. The same distinction applies here. A non-metric score can be a valid measure of a well-named attribute, such as mass on an answer set. A true metric can still be the wrong attribute for the decision, for example Hellinger distance when only known-answer mass is required, or Uhlmann fidelity when only shots are available. Clipping a score at zero, as chance-corrected success does below the chance baseline, is a representation failure on that region: every worse-than-chance job is assigned the same number.

The Goal–Question–Metric paradigm [1], [9] separates validating a *measure* from validating a *prediction system*. ESP is closer to the latter: it predicts hardware success from calibration data and is not a measure of the observed histogram. Mohapatra *et al.* [10] validate HF, TVD, ESP, and PST as proxies of density-matrix fidelity. That work validates a prediction system. It does not decide whether the number matches the attribute chosen for a pass/fail decision.

III. RELATED WORK

Gilchrist, Langford, and Nielsen [2] give criteria a quantum information-processing distance must meet and recommend theoretically and experimentally meaningful distances. Luo and Zhang [3] and Ma *et al.* [4] develop Hellinger and affinity distances and fidelity-induced metrics on state space. Uhlmann [11] and Nielsen and Chuang [5] provide the textbook definitions. Those papers settle the fidelity-versus-distance question. They do not apply representational measurement theory to job-level pass/fail decisions.

Mohapatra *et al.* [10] correlate HF, TVD, ESP, and PST with density-matrix fidelity (ESP Spearman 0.88; PST Pearson 0.92). That work validates proxies of a tomography-based quantity. The present paper asks which number matches the attribute named before any correlation is computed.

Hacaloglu *et al.* [12] review quantum software size: lines of code, COSMIC, qubits, gates, and cyclomatic complexity. The questions of what, how, and why to measure are the correct software-engineering questions applied to a different attribute. Size measurement can be rigorous while execution fidelity remains unclassified.

Testing-oracle surveys [13], [14] already use TVD, Hellinger distance, and χ^2 as pass/fail checks. They do not

TABLE I: Sampler and Estimator interpretations of a job.

	Sampler	Estimator
Kept	$\hat{P}$, or mass on a goal set E	estimates $\langle \widehat{O_i} \rangle$
Discarded	observable algebra; most phase	the histogram as a histogram
Decision	Did $\hat{P}$ match E or P^* ?	Did the correlators match?

TABLE II: Estimator quantities derived from the same shots.

Quantity	Typical formula	Kind
Expectation $\hat{e}v$	job evs	estimate
Success probability	$(1 + \hat{e}v)/2$	score
Observable fidelity	$1 - \hat{e}v - \text{ideal} /2$	similarity
Relative error	$ \Delta ev / \max(\text{ideal} , \epsilon)$	score
SNR	$ \hat{e}v /\text{std}$	stability
Depolarization	mean of $\hat{e}v_i/\text{ideal}_i$	model fit

ask whether those oracles are metrics in the M1–M4 sense or valid Fenton measures of the chosen attribute. Studies of circuit understandability and Q-COSMIC functional size likewise measure the product, not the execution.

This paper does not claim that the M1–M4 distinction is new, and it does not claim that fidelity is not a distance as a new result. The gap addressed here is a representation-and-GQM treatment of the job-level decision to mark an execution successful, with Sampler and Estimator as the first distinction and with explicit reference requirements. If later work occupies that cell, the analytical results and the DSR case study remain to be argued on their own terms.

IV. EXECUTION MODEL

A shot is a bitstring $x \in \{0, 1\}^m$. A job is a finite multiset of shots. The primitive chooses the empirical relation that will be preserved (Table I).

A. Sampler and Estimator Interpretations

On the Sampler interpretation the derived objects are the histogram $\mathcal{C}$, $\hat{P} = \mathcal{C}/S$, a goal set E , an optional full ideal P^* , and optional inverse-circuit PST. Typical tasks in the evaluation corpus are Grover search, quantum Fourier transform (QFT) round-trips, Bernstein–Vazirani (BV), and teleportation.

On the Estimator interpretation the same shots are contracted. For a Pauli observable O with eigenvalues in $\{\pm 1\}$,

$$\langle \widehat{O} \rangle = \frac{1}{S} \sum_{s=1}^S \lambda(x_s), \quad p_{\text{succ}} = \frac{1 + \langle \widehat{O} \rangle}{2}. \quad (3)$$

Table II lists the usual derived Estimator quantities. None of them is a metric on the histogram simplex. The absolute deviation $|\hat{e}v - \text{ideal}|$ is a metric on the value line. Variational and chemistry workloads use this interpretation.

The following assignments are inconsistent with the representation condition. Hellinger fidelity is not defined for an Estimator job whose result contains only evs and stds . DSR is not defined for an Estimator job unless the raw counts are reconstructed and an E is supplied, which returns the attribute

TABLE III: Reference classes for job-level quantities.

Class	Required input	Unavailable when
R1 known E	answer bitstrings	no discrete answers
R2 full P^*	ideal on $\{0, 1\}^m$	m exceeds simulation capacity
R3 ideals $\langle O_i \rangle$	expectations or a Hamiltonian	only counts are retained
R4 tomography or uncompute	ideal state or $U^\dagger U$	tomography cost or doubled depth

to the Sampler interpretation. The quantity $(1 + \hat{e}v)/2$ is not Sampler p_E . SNR is not a success score: a large SNR can occur for an incorrect expectation.

B. Reference Information

A formula can be a valid metric and still be inapplicable. The missing requirement is the reference information. Table III lists four reference classes used in the rest of the paper.

R1 quantities include p_E , chance-corrected success, coarse TVD and Hellinger distance, and Michelson DSR. R2 quantities include Hellinger distance, F_H , TVD, and $1 - \text{TVD}$. R3 quantities include observable fidelity, $|\Delta ev|$, and relative error. R4 quantities include Uhlmann and density-matrix fidelity and PST.

The first question in Section VIII-A chooses the attribute (Sampler versus Estimator). R1–R4 determine whether that attribute is computable. Mixing a Sampler quantity onto Estimator evs , or treating $(1 + \hat{e}v)/2$ as Sampler p_E , violates the representation condition.

V. CLASSIFICATION OF EXECUTION QUANTITIES

Table IV classifies common job-level quantities. Scale types are tentative where a full representation proof is not restated; the last column records the principal limitation.

Hellinger distance and TVD inherit M1–M4 as the L^2 and L^1 metrics on the probability simplex. Coarse TVD and coarse Hellinger distance inherit the same axioms on the push-forward simplex Δ_{K+1} that keeps each $x \in E$ and lumps the complement into one bin.

Proposition 1 (Coarse quantities at $K = 1$): Let $K = |E| = 1$ and write p_E for the observed mass on the single expected string. On the coarse simplex the observed histogram is $(p_E, 1 - p_E)$ and the ideal is $(1, 0)$. Then the coarse total variation distance equals $1 - p_E$ and the coarse Hellinger fidelity equals p_E .

Proof: $\text{TVD} = \frac{1}{2}(|p_E - 1| + |(1 - p_E) - 0|) = 1 - p_E$. The Bhattacharyya coefficient is $\text{BC} = \sqrt{p_E \cdot 1} + \sqrt{(1 - p_E) \cdot 0} = \sqrt{p_E}$, so $F_H = \text{BC}^2 = p_E$. ■

The identities are exact for the coarse construction. They do not identify coarse HF with full-histogram HF unless the true ideal is a delta on E . Finite-iteration Grover has residual amplitude outside E ; QFT round-trip and teleportation ideals are such deltas. Those are different attributes (R1 versus R2), not a contradiction of Proposition 1.

TABLE IV: Classification of common job-level quantities. A metric satisfies M1–M4 on the named space. Similarities and scores are not candidates for that test.

Quantity	Attribute	Ref.	Kind	Limitation
Success rate p_E	mass on E	R1	score	ignores competing peaks; not injective
Chance-corrected success	above-chance mass on E	R1	clipped score	censors $p_E < b$; $K = 2^m$ forces 1
Hellinger distance H	histogram distance	R2	metric (L^2)	needs P^* ; $O(2^m)$
Hellinger fidelity F_H	histogram overlap	R2	similarity	same per-job information as H ; means differ
TVD	L^1 histogram distance	R2	metric	needs P^* ; phase-blind
$1 - \text{TVD}$	histogram similarity	R2	similarity	compose only after mapping back to TVD
Uhlmann / DM fidelity	state overlap	R4	similarity	tomography; not a bitstring job tool
Bures / sine distances	state distance	R4	metrics	same cost as the fidelity they complete
ESP	predicted hardware success	—	prediction	independent-error model; not observed $\hat{P}$
PST	return-to-zero of $U^\dagger U$	R4	score	different program; T1 bias on $ 0\rangle^{\otimes n}$
Coarse TVD	distance on Δ_{K+1}	R1	metric	forgets structure inside the complement
Coarse HF	coarse overlap	R1	similarity	equals p_E at $K = 1$
Michelson DSR	mean expected peak vs. max competitor	R1	clipped score	mean-versus-max; T1 bias; K -dependence
$ \hat{e}v - \text{ideal} $	expectation deviation	R3	metric	one observable; ignores histogram shape
Observable fidelity	expectation similarity	R3	similarity	mean over observables is a new measure
$(1 + \hat{e}v)/2$	+1-eigenvalue mass of a Pauli	R3	score	not Sampler p_E
Relative error	relative deviation	R3	score	unstable near 0
SNR	estimate stability	—	derived	can be large on the wrong value

VI. ANALYTICAL RESULTS

A. Threshold Invariance

Lemma 1 (Threshold invariance): Let $d(\cdot, Q)$ be a real-valued quantity on histograms with fixed reference Q , and let ϕ be strictly monotone. For a threshold τ on d , define $\tau' = \phi(\tau)$. Then $d(P, Q) \leq \tau$ if and only if $\phi(d(P, Q))$ stands in the corresponding relation to τ' ($\leq$ if ϕ is increasing, $\geq$ if ϕ is decreasing).

Proof: Fix Q . The map $P \mapsto d(P, Q)$ is a real number. A strictly increasing ϕ preserves order, so $d(P, Q) \leq \tau \Leftrightarrow \phi(d(P, Q)) \leq \phi(\tau)$. A strictly decreasing ϕ reverses order, so $d(P, Q) \leq \tau \Leftrightarrow \phi(d(P, Q)) \geq \phi(\tau)$. In either case the converted threshold $\tau' = \phi(\tau)$ yields the same partition of jobs. ■

Single-job pass/fail is therefore invariant inside a strictly monotone class after threshold conversion. The classes used later are:

- **Hellinger class.** Hellinger distance H , Bhattacharyya coefficient BC , Hellinger fidelity $F_H = BC^2 = (1 - H^2)^2$, and the angle $\theta = \arccos BC$ are strictly monotone transforms of BC .
- **TVD class.** TVD and $1 - \text{TVD}$ are exact complements.
- **Known-answer class,** $K = 1$. p_E , coarse TVD similarity, and coarse HF are identical. Chance-corrected success is a clipped affine transform of p_E .
- **Expectation class.** $|\hat{e}v - \text{ideal}|$ and observable fidelity $1 - |\Delta ev|/2$ are complementary up to the factor $1/2$.

B. Aggregation

Let $\phi(H) = (1 - H^2)^2$. Arithmetic means need not preserve a ranking obtained from H , because ϕ is nonlinear. A constructed pair shows the mechanism. Take three Hellinger distances $A = \{0.10, 0.10, 0.90\}$ and $B = \{0.40, 0.40, 0.40\}$. Then $\text{mean}(H_A) = 0.367 < 0.400 = \text{mean}(H_B)$, so A is the better group if lower mean distance is the criterion. The corresponding fidelities are $\phi(0.10) = 0.980$ and $\phi(0.90) = 0.036$,

so $\text{mean}(F_H(A)) = 0.665 < 0.706 = \text{mean}(F_H(B))$: B is the better group if higher mean fidelity is the criterion. Median aggregation would preserve order inside a strictly monotone class (0.10 versus 0.40). Whether existing QPU rows invert under mean H versus mean F_H is examined in Section VII-E; that analysis has not yet been run. Until it is, this paper does not claim a provider ranking flip on the corpus. The constructed example shows that the operation can invert a ranking; it does not show that the corpus realises the inversion.

C. Composition

If d satisfies the triangle inequality on a named space X , then $d(P, R) \leq d(P, Q) + d(Q, R)$. That statement is undefined for a similarity s . The induced dissimilarity $1 - s$ must be examined separately; it need not be a metric. For TVD similarity $s = 1 - \text{TVD}$, the corresponding bound is $s(P, R) \geq s(P, Q) + s(Q, R) - 1$. From $\text{TVD}(P, R) \leq \text{TVD}(P, Q) + \text{TVD}(Q, R)$ and $s = 1 - \text{TVD}$ one obtains $1 - s(P, R) \leq (1 - s(P, Q)) + (1 - s(Q, R))$, hence the same inequality. The bound is weaker than a triangle inequality and can be negative on the right-hand side. Errors that must be composed should be converted to a distance first [2]. The same restriction applies to averaging observable fidelities across Pauli terms: the mean of $1 - |\Delta ev_i|/2$ is not the transform of the mean deviation.

D. Scale Type

A number in $[0, 1]$ is not automatically a ratio scale. Chance-corrected success clipped at 0 does not represent “worse than chance.” A cutoff such as 0.8 is meaningful only after the attribute, the chance baseline, and the scale type are named [1], [6]. The following statements are therefore different kinds of claim:

- “Job A has higher p_E than job B ” is an ordinal statement on a shared E .

TABLE V: Sampler example ($K = 1$).

Quantity	Value	Kind
p_E	0.40	score
CCS	$\text{clip}((0.40 - 0.25)/0.75) = 0.20$	clipped score
Coarse observed / ideal	$(0.40, 0.60)$ vs. $(1, 0)$	push-forward
Coarse TVD	0.60	metric on Δ_2
Coarse TVD similarity	$0.40 = p_E$	$K = 1$ identity
Coarse HF	$0.40 = p_E$	$K = 1$ identity
Coarse Hellinger distance	$\sqrt{1} - \sqrt{0.40}$	metric on Δ_2
Michelson DSR	$1/3$	contrast 0.40 vs. 0.20

- “The mean p_E over a batch is 0.41” is a frequency statement and is meaningful as an average of proportions.
- “ $F_H = 0.80$ is twice as good as $F_H = 0.40$ ” is a ratio statement that the similarity scale does not licence.
- “Pass if $F_H \geq 0.8$ on every algorithm” reuses a cutoff across tasks that need not share a reference or a chance baseline.

VII. EVALUATION

No new QPU jobs are introduced. The evaluation uses existing rows and two hand calculations. A quantity is reported only if it demonstrates invariance, a change of conclusion, or an unavailable reference.

A. Experimental Setup

The corpus consists of previously executed Sampler jobs used for known-answer evaluation, including Grover search, QFT round-trips, Bernstein–Vazirani, and teleportation, together with one Estimator scalar example. Full-histogram Hellinger and TVD values are used where an ideal distribution P^* is available. Known-answer scores are used where only E is available.

B. Sampler Example

Consider counts $\mathcal{C}(01) = 40$ and $\mathcal{C}(00) = \mathcal{C}(10) = \mathcal{C}(11) = 20$, with $E = \{01\}$, $S = 100$, and $m = 2$, so $K = 1$ and chance $b = 1/4$ (Table V).

The decision “pass if coarse TVD ≤ 0.30 ” is the same as “pass if coarse TVD similarity ≥ 0.70 .” That identity is the content of Lemma 1. The four coarse numbers that equal p_E or $1 - p_E$ are not independent evidence.

C. Estimator Example

Let the ideal be $\langle ZZZZ \rangle = 1$, the estimate $\hat{e}v = 0.70$, and the standard error 0.04 (Table VI). The decision “pass if $|\hat{e}v - \text{ideal}| \leq 0.20$ ” is the same single-observable decision as “pass if observable fidelity ≥ 0.90 .” Averaging fidelities across several observables is not the transform of the mean deviation. SNR is not a success score.

TABLE VI: Estimator example.

Quantity	Value	Kind
Ideal $\langle ZZZZ \rangle$	1	R3 reference
$\hat{e}v$	0.70	estimate
Success probability	0.85	not Sampler p_E
Observable fidelity	0.85	similarity
Relative error	0.30	score
SNR	17.5	stability

D. Single-Job Invariance

On existing rows that store both Hellinger and TVD pairs, ranking by H should reverse ranking by F_H , and ranking by TVD should reverse ranking by $1 - \text{TVD}$. If there are no ties, Spearman $\rho = -1$ inside each pair. That result is the empirical counterpart of Lemma 1. It is not a claim that coarse HF matches full HF. Finite-iteration Grover has residual amplitude outside E ; QFT round-trip and teleportation ideals are deltas on E . Those are different attributes (R1 versus R2).

E. Aggregation of Distances and Similarities

The protocol is to group existing jobs by provider and workload, compute the mean and median of H and of F_H (and of the TVD pair), and test whether any provider or workload ranking inverts. A ranking inversion would show that the choice of quantity changes a multi-job conclusion. Absence of inversion is a negative result and restricts the empirical claim to composition, scale, and reference requirements.

This analysis has not been run. The paper therefore does not assert an IBM-versus-Rigetti ranking inversion under mean HF versus mean Hellinger distance.

F. Unavailable Reference Distributions

Bernstein–Vazirani and other broad-ideal circuits in the existing corpus make P^* incomputable at the widths of interest, while a known E remains available. Full HF and TVD cannot be computed. Known-answer scores and coarse-space distances remain. That is an applicability result, not a performance comparison. A quantity that is a metric on paper is still inapplicable if its reference class cannot be obtained. The same restriction applies to Uhlmann fidelity when tomography is not performed: the formula remains a similarity on density operators, and the job remains unevaluable under R4.

G. Teleportation Results

On existing teleportation jobs, IBM median success and chance-corrected success are lower than Rigetti at payload sizes 3 and 4 (Table VII). Michelson DSR medians are 0 for both providers at payloads 2 and 4. Amplitude damping concentrates mass on the all-zero string and inflates the strongest competitor, so the mean-versus-max contrast saturates while mass on E still separates providers. This is a change of conclusion inside R1 and does not depend on the aggregation analysis of Section VII-E.

TABLE VII: Teleportation median success rate. Michelson DSR medians are 0 for both providers at payloads 2 and 4.

Payload	IBM	Rigetti	δ
1	0.615	0.700	-0.50
2	0.251	0.400	-0.34
3	0.132	0.400	-0.82
4	0.066	0.200	-0.76

TABLE VIII: Questions and recommended quantities.

Question	Quantity
How far is $\hat{P}$ from P^* ?	TVD or H on Δ_{2^m} (R2)
How far is the coarse histogram from E ?	coarse TVD or coarse H (R1)
Did enough shots land in E ?	p_E ; CCS if b is not negligible
Do expected peaks dominate competitors?	Michelson DSR (state T1 bias and K)
Is P^* computable?	if not, only R1 remains
How far is $\hat{e}v$ from the ideal?	$ \Delta_{ev} $ or observable fidelity (R3)
Is the estimate stable enough to decide?	SNR and shot budget, not a success score
Are several observables aggregated?	name the scale; do not average fidelities as the transform of the mean deviation

VIII. DISCUSSION

A. Selection of a Quantity

The goal is to decide whether a quantum execution succeeded. The first question is whether the shots are interpreted as a histogram or known answer, or as estimates of observables. Table VIII maps subsequent questions to quantities.

On the Sampler interpretation, histogram distance should be measured with TVD or Hellinger distance on the space that matches the reference: Δ_{2^m} if P^* is available (R2), Δ_{K+1} if only E is available (R1). Mass on E should be reported as success rate, or as chance-corrected success when $K/2^m$ is not negligible, with the clip stated if values below chance are mapped to 0. Michelson DSR answers a peak-dominance question and is not independent confirmation of p_E when $K = 1$. If P^* is not computable, only R1 quantities remain.

On the Estimator interpretation, deviation from the ideal should be reported as $|\hat{e}v - \text{ideal}|$ or as observable fidelity, with a converted threshold if both are reported. SNR and shot budget answer precision, not task success. If several observables are aggregated, the scale must be named.

Variational workloads use the Estimator interpretation. DSR does not apply unless an E is defined. Thresholds require shots, a chance or ideal baseline, and a stated scale type. A single cutoff such as 0.8 is not a meaningful operation on every scale.

B. Case Study: Differential Success Rate

The Differential Success Rate profile is not a metric. It is a known-answer evaluation record that mixes unary scores with two distances on a coarse simplex. Write S for the shot count,

TABLE IX: DSR profile at $K = 1$.

Component	Value
Success rate	p_E
Coarse TVD	$1 - p_E$
Coarse TVD similarity	p_E
Coarse HF	p_E
Coarse Hellinger distance	$\sqrt{1 - \sqrt{p_E}}$
Chance-corrected success	$\text{clip}((p_E - b)/(1 - b), 0, 1)$

E for the expected set, $K = |E|$, and $b = K/2^m$ for the chance baseline. The computed components are

$$p_E = \frac{1}{S} \sum_{x \in E} \mathcal{C}(x), \quad (4)$$

$$\text{CCS} = \text{clip}\left(\frac{p_E - b}{1 - b}, 0, 1\right), \quad (5)$$

$$\text{coarse-TVD} = \frac{1}{2} \sum_k |p_k - q_k|, \quad (6)$$

$$\text{DSR}_{\text{Mich}} = \text{clip}\left(\frac{\bar{p}_E - p_{\text{comp}}}{\bar{p}_E + p_{\text{comp}}}, 0, 1\right), \quad (7)$$

where the coarse TVD is evaluated on the push-forward that keeps each $x \in E$ and one complementary bin, $\bar{p}_E = p_E/K$, and $p_{\text{comp}} = \max_{x \notin E} p_x$. Coarse Hellinger distance and coarse HF are the same functions on that smaller simplex.

Table IX records the $K = 1$ identities that follow from Proposition 1.

The limitations of the profile are as follows.

- 1) **$K = 1$ degeneracy.** When $|E| = 1$, the non-Michelson profile has one degree of freedom, p_E (Table IX). Reporting four of those numbers as independent evidence overstates the information.
- 2) **Clipping.** Chance-corrected success maps every job with $p_E \leq K/2^m$ to 0. The order “worse than chance” is not represented.
- 3) **Mean versus maximum.** Michelson DSR compares the mean expected peak to the strongest competitor and then clips below zero. Dividing by K changes the attribute even when p_E is unchanged. Jobs with different K are not on the same scale.
- 4) **Known E .** The profile never constructs an object of size 2^m . That is both its information advantage and its applicability restriction.
- 5) **Phase.** Bitstring mass does not represent relative phase.
- 6) **Relaxation.** Amplitude damping concentrates mass on the all-zero string. When that string is the strongest competitor, Michelson DSR collapses (Section VII-G).
- 7) **Variational workloads.** Without a discrete E , the profile is undefined. Estimator jobs that expose only `evs` and `stds` are out of scope unless counts are reconstructed and an E is supplied.

The push-forward onto $\{E\} \cup \{\text{other}\}$ makes coarse TVD and coarse Hellinger distance metrics on Δ_{K+1} . Success rate is a valid score of mass on E . The profile is the record

$$(p_E, \text{CCS}, d_{\text{TV}}^{\text{coarse}}, d_H^{\text{coarse}}, \text{optional Michelson}) \quad (8)$$

plus the derived similarities. Averaging those entries into one number would invent a fifth derived measure with no attribute [1].

The profile is applicable when the task has an analytically known answer set, the decision is task-level rather than state-level or observable-level, the primitive is Sampler, and P^* may be incomputable. It is not a replacement for TVD or H on Δ_2^m , for Uhlmann fidelity, or for Estimator deviation. The claim is not that DSR outperforms HF. The claim is that it measures a different attribute and that several of its reported numbers fail representation or independence tests.

IX. THREATS TO VALIDITY

a) *Internal validity.*: Finite shots make every derived number an estimate. Thresholds that ignore binomial uncertainty overstate precision. The corpus mixes providers, dates, and workloads; calibration drift and queue conditions are not controlled.

b) *Construct validity.*: Section VII-F treats the simulation limit as missing R2 information, not as a missing implementation. Grover ideals are not deltas; identifying coarse HF with full HF on those jobs would be a construct error. The mean-inversion example in Section VI-B is constructed. The aggregation protocol in Section VII-E has not been executed. Until it is, the strongest observed change of conclusion in the corpus is the teleportation contrast between chance-corrected success and Michelson DSR (Section VII-G).

c) *External validity.*: The evaluation reuses an existing known-answer corpus. Results may not extend to Estimator-only variational workloads or to jobs for which no E can be defined.

d) *Bias.*: DSR is implemented in the same library that motivated the case study. The case study is therefore written to state limitations before the applicable scope.

e) *Scope.*: The paper does not propose a new distance on density operators and does not claim quantum advantage. It does not classify internal product metrics (depth, two-qubit count) except to exclude them as execution-success measures.

X. CONCLUSION

The limitation of execution “metrics” is not the popularity of the word. The same label is attached to distances, similarities, unary scores, prediction systems, and Estimator scalars that encode different attributes, require different references, and permit different operations.

For one job, one fixed reference, and a converted threshold, monotone pairs agree. Distinguishing fidelity from distance does not, by itself, change a pass/fail decision. Conclusions can still change when jobs are aggregated, when errors are composed, when ratio cutoffs are reused across tasks, when P^* or an ideal expectation is unavailable, and when Sampler formulas are applied to Estimator results.

The practical rule is the same as in software measurement. Choose the attribute and the decision first. Then choose a quantity whose domain, reference class, scale, and permitted operations fit that decision. Reserve *distance metric* for M1–M4 on a named space. Call everything else a measure, score,

or similarity only after those pieces are named. A non-metric score can be the right measure. A true metric can still be the wrong one.

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